

End-to-End Customer Behavior Analysis

Project Documentation & Pipeline Case Study

1. Project Overview & Objectives

In modern retail enterprise environments, understanding the intersection of demographic variance, channel logistics, and marketing promotion strategies is essential for driving net margin improvements. This project builds a fully realized data pipeline to answer key strategic questions:

- **Demographic Insights:** Assessing how gender and age cohorts influence purchase metrics and item categorical distributions.
- **Promotion Performance:** Evaluating whether discount campaigns cultivate profitable customer lifetimes or generate unsustainable margin dilution.
- **Logistics Optimization:** Identifying operational fulfillments that demonstrate higher check-out sizes and locating operational bottlenecks.

[Raw CSV Data] —> [Jupyter Notebook (Python)] —> [MySQL Database] —> [Power BI Report]

2. Technical Architecture

Phase	Infrastructure / Tool	Core Operations & Methodologies
Data Engineering	Python (Jupyter Notebook)	Data Profiling, Missing Value Imputation, Type Casting, Format Standardization via Pandas & NumPy.
Database Layer	MySQL Workbench	Relational Schema Ingestion, Aggregations, Sub-queries, CTEs, Window Functions (ROW_NUMBER over Partitioning).
Business Intelligence	Power BI Desktop	Star-schema Data Modeling, Matrix DAX Expressions, Dynamic Visual Interactivity & Drill-downs.

3. Pipeline Phase Execution Details

Phase 1: Data Preprocessing & Cleaning (Python)

The raw transaction logs were audited using `Customer_Behaviour.ipynb`. Structural preparation involved isolating null values, parsing continuous real-number formats out of corrupted string identifiers (e.g., stripping special characters from numerical representations), and converting target categorical flags into uniform structural states (e.g., resolving case inconsistencies across binary boundaries).

Phase 2: Relational Database Analytics (MySQL)

The clean dataset was structured into a transactional table mapping to `addydb1.mytable`. Analytical scripts in `customer_behavior_Sql.sql` solved complex business aggregation constraints:

- Calculated total segmented cohorts via group constraints, filtering high-value contributors whose behavior outpaced demographic means using active nested subqueries.
- Constructed structured analytical reporting layers using Common Table Expressions (CTEs) combined with partition-based analytical window functions:

```
ROW_NUMBER() OVER (PARTITION BY category ORDER BY count(customer_id) DESC)
```
- Dynamically extracted the exact top three leading item products across all organizational marketplace lines.

Phase 3: Dashboard Development & UI/UX (Power BI)

The relational tables were linked into an optimized Power BI data engine model inside `customer_behavior_pbi.pbix`. Time-series metrics and dynamic check-out sizes were mapped out cleanly via targeted DAX functional measures, forming a robust dashboard canvas tailored for corporate performance evaluation.

4. Strategic Business Insights

• **Discount Elasticity Paradox:** While discount activations systematically prompt elevated volume and transaction velocities, deeper cohort sub-query metrics confirm that premium, organic non-discount purchase paths retain a significantly higher baseline Average Order Value (AOV).

• **Category Performance Distribution:** Leveraging window partitions revealed that over 40% of standard category margins are localized within only a small subset of core catalog options. This establishes prime operational target lines for cross-selling strategies.

• **Fulfillments & Shipping Logistics:** Data points associated with Express checkout alternatives consistently yield higher net unit volumes per checkout, validating the business case for introducing value-add express tier perks directly tied to high minimum-spend baskets.

Documentation compiled for GitHub Project Portfolio Showcase. Verified across all development phases.